OKEANOS

SMART WATER TECHNOLOGY

OKEANOS SMART WATER PLATFORM OVERVIEW

OKEANOS SMART WATER OY URAKOITSIJANTIE 15, 064500 PORVOO, FINLAND

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INTRODUCTION

Atmospheric Water Generators or AWGs are a proven technology that extract moisture from the air to provide a supply of clean, safe drinking water. There are several techniques for extracting water using chemical desiccants and passive heat exchangers, however, currently most of these techniques are not particularly efficient and are difficult to scale.

The most reliable and high-yielding technique is electro-mechanical extraction. Essentially, machines using this technique are very similar to the dehumidifiers and air conditioning units used in domestic and commercial buildings. The electro-mechanical, “AWG”, is a specialised version of a dehumidifier that is optimised to extract the maximum amount of water from the available moisture in the air. The components used are all “food-grade” so that the water is not contaminated and most of these systems include sophisticated filtration systems and UV treatment technology to produce a final product that is safe, clean and contaminant-free water fit for consumption..

In the right environmental conditions, these AWG machines can extract up to 75% of the available moisture from the air passing through the system. These machines have successfully been used for many years in small-scale water production for off-grid hotels and small drinking water dispensaries. Their application has been proved to be highly beneficial and efficient, in regards to the small-scale operations where they have been previously utilized.

For any of the technologies to be deployed and produce water, the yield will be dependent on the environmental conditions of the location it is placed and operated. The most important factor is the amount of water in the atmosphere at the location. For electromechanical systems to work, it is necessary to condense the moisture in the air. This is achieved by cooling the air that is passed through the AWG down to a specific “dew point”. At the required dew point, the water will condense onto a coil and can then be collected, filtered and stored ready for consumption. The dew point will depend on the relative humidity and the ambient temperature of the location. If the ambient temperature is too low, then the dew point will be close to the freezing point and this can cause problems with the condenser freezing and result in a yield of low water. Therefore, the placement of the AWG machine is fundamental to the efficiency and final output of the system.

Due to the variations in ambient temperature and humidity, planning and understanding the characteristics of a specific location is key to making AWG deployments reliable and scalable. Okeanos can modify the machines to perform better at low temperatures, but in general, the machines work more efficiently in higher temperature and high humidity climates. Information pertaining to a specific location is fundamental to effective planning and management of a water production system or plant.

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OKEANOS AWG PLATFORM COMPONENTS

LOCATION PLANNING TOOL

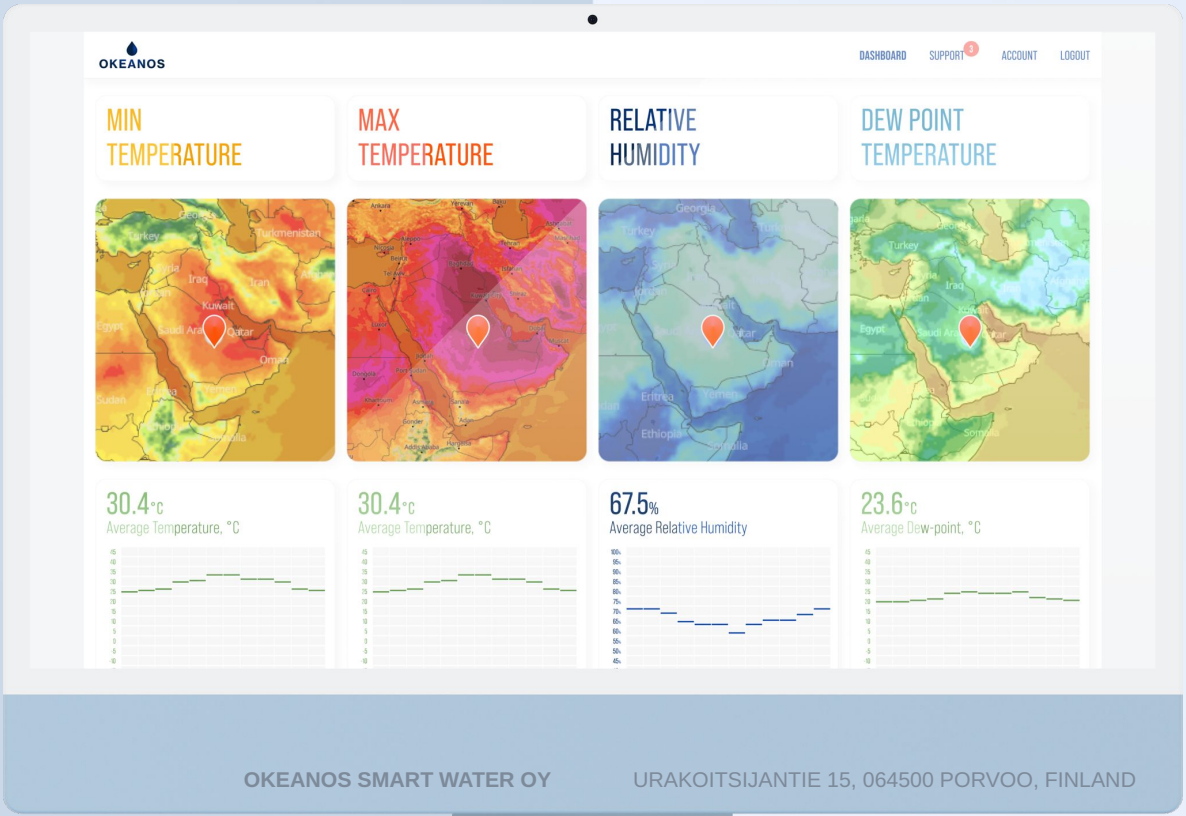
MANAGEMENT AND CONTROL SYSTEM

BILLING AND EXTERNAL DATA INTEGRATIONS

OKEANOS LOCATION PLANNING TOOL

The environment can vary greatly depending on the location's characteristics. Even in locations 5 miles apart, there can be large differences in temperature and humidity. Altitude and air pressure also play an important role in determining efficiency and effectiveness of a system or plant. Proximity to bodies of standing water, lakes and the coast can greatly affect temperature and humidity. Prevailing winds can make a difference as well as vegetation and forests, etc. The micro-climate that the AWG is located in can make a big difference to its effectiveness in producing water.

Using historical weather data, it is possible to calculate a guide for the projected yield of water in different locations. The actual yield will of course depend on the "real" weather conditions in the location, that may vary from year to year. This data gives us a guide to the likely performance and projected result in the specific location. The Okeanos platform incorporates historic weather data to allow us to calculate the likely production yield for any location. This unique capability means that we can predict the water yield and calculate how many AWG machines are required to meet the customer's requirements. In cases where there are low yield months, we can specify how much storage capacity is required to ensure a year-long water supply is available or how many machines are necessary to reach the intended production output.



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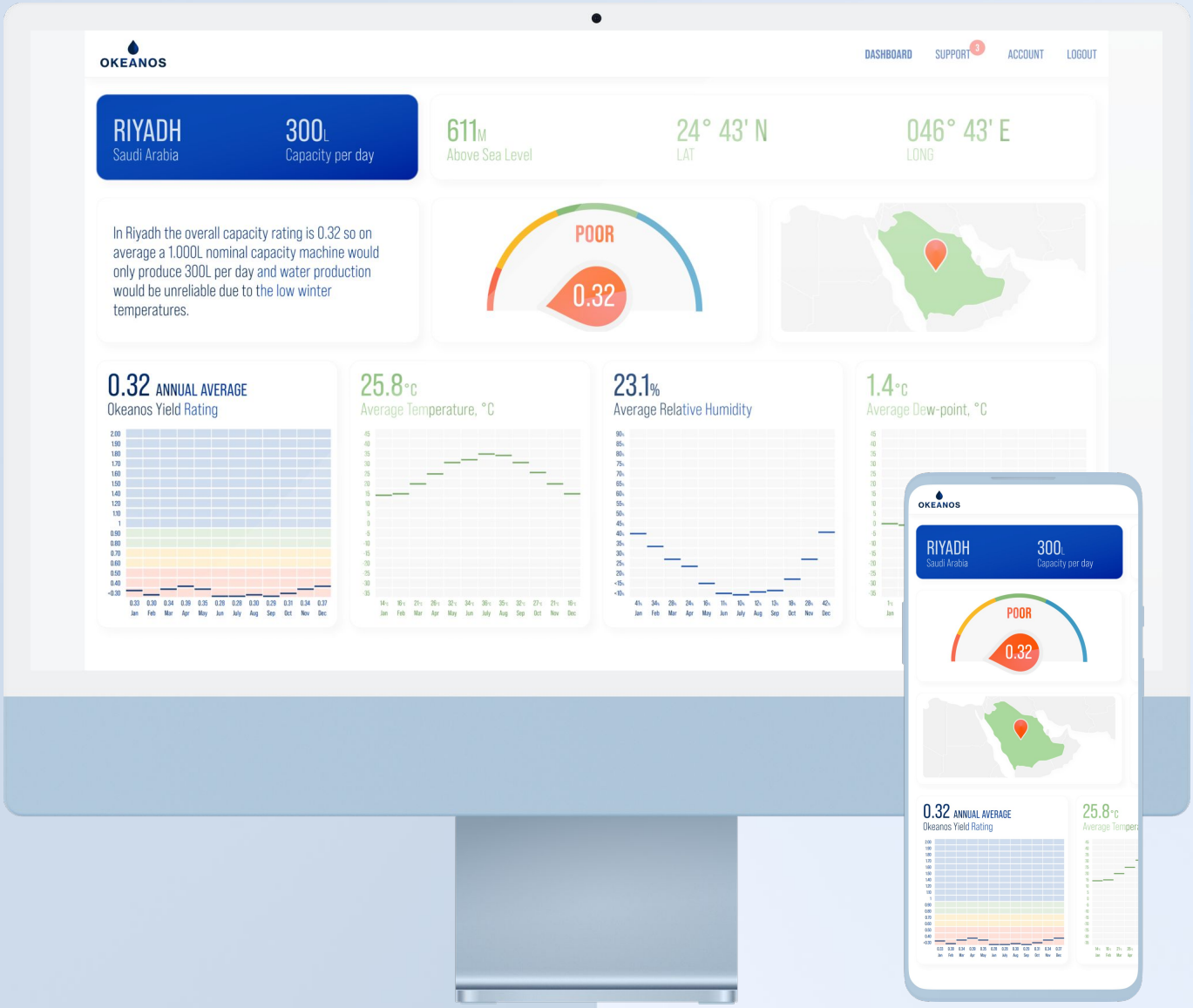
In the examples below, we can see that in three (3) cities in Saudi Arabia, there are huge differences in the potential yield for water production.

RIYADH



In Riyadh the water production index is 0.32, so on average a 1000L nominal capacity machine would only produce 300L per day and water production would be unreliable due to low winter temperatures. The viability of the site would be improved by a low temperature variant of the machine, however, due to the low humidity, the yield would likely be poor.

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JEDDAH

Jeddah has an overall rating of 0.99 so on average a 1000L nominal capacity machine will produce just under 1000L per day, the yield is good during 6 months of the year and with some storage a reliable water production system could be built.

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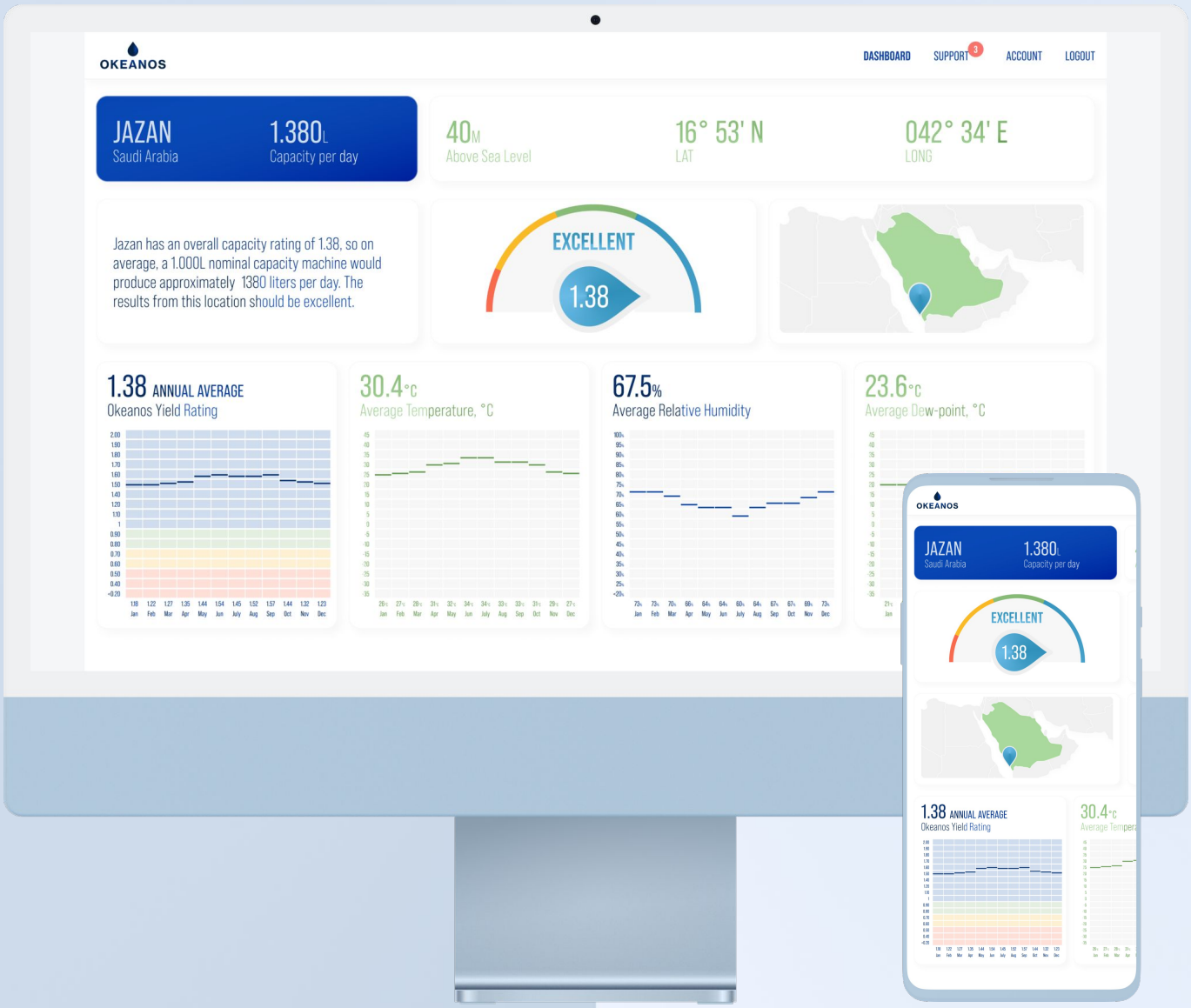
JAZAN



Jazan has an overall rating of 1.38 so on average a 1000L nominal capacity machine would produce approx 1380 L per day, the results from this location should be excellent.

As you can see in the 3 sites above, the yield and the viability of the sites can vary enormously. This shows how important it is to make a plan for the location before deploying and operating the machines. The Okeanos platform location planning tool will provide an excellent guide to pinpoint locations with the most potential for positive results.

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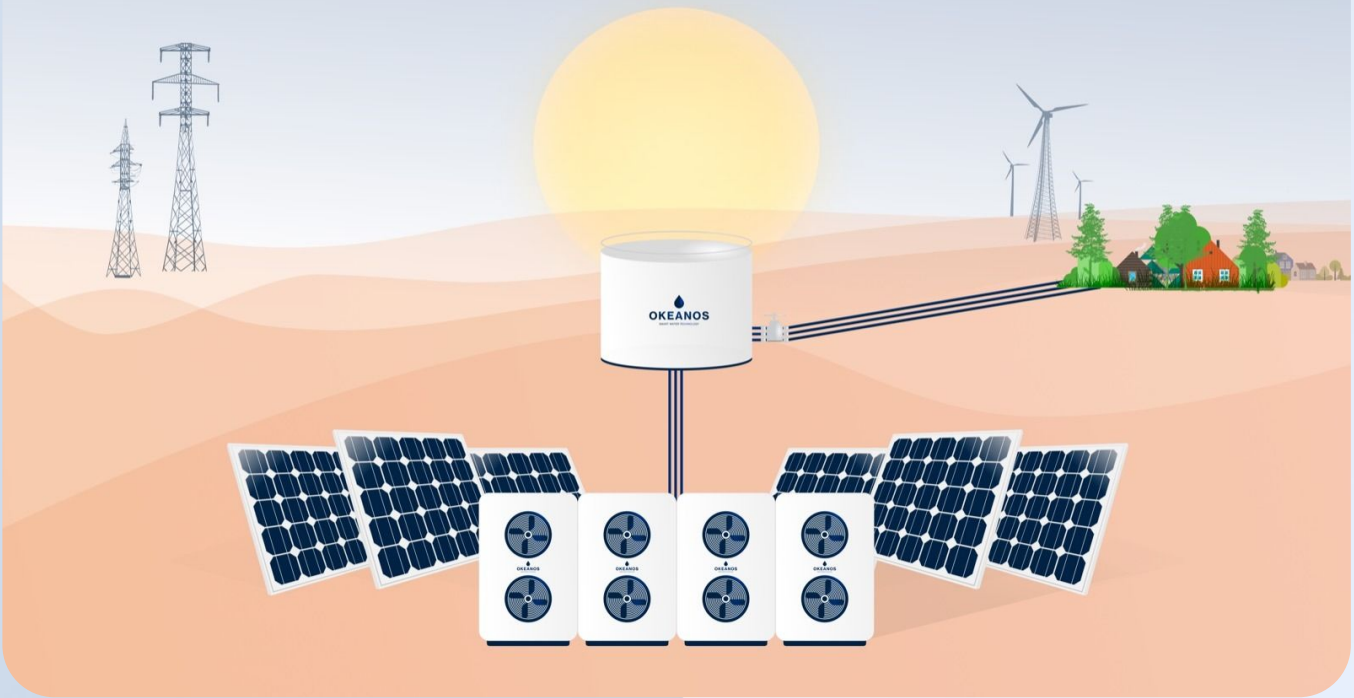
MANAGEMENT AND CONTROL CAPABILITIES

AWG machines can be deployed in many different configurations, from one-off deployments with single machines for single dwellings or in larger capacities to supply small communities. To scale capacity further, the most flexible method is to build solutions by aggregating and connecting multiple machines into a “water farm”. Using multiple machines together has several advantages.

Resilience - Redundant capacity will make for a highly reliable system.
Flexible footprint - machines can be configured in different configurations to maximise yield and use of space.
Ease of maintenance
Easy to transport



CONTROL

In order to manage a machine or group of machines, Okeanos makes use of process control protocols to control the AWG machines. Using industry standard protocols we can connect the machines together in a network and manage them as a system.

All machines supplied by Okeanos include our, “smart water control system”, the control system allows us to remotely control and monitor individual AWG machines or complete water farms.

Our cloud-based control system continuously collects data from all of the AWG machines via a resilient satellite and cellular IOT system.

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We have developed management dashboards making it easy to visualise all aspects and parameters. Alarms and alerts can for potential faults and anomalies help to keep the systems running smoothly with preventive maintenance. We can also visualise the water production statistics for the water farm as a whole or for individual machines.

Some example dashboards are shown below:-

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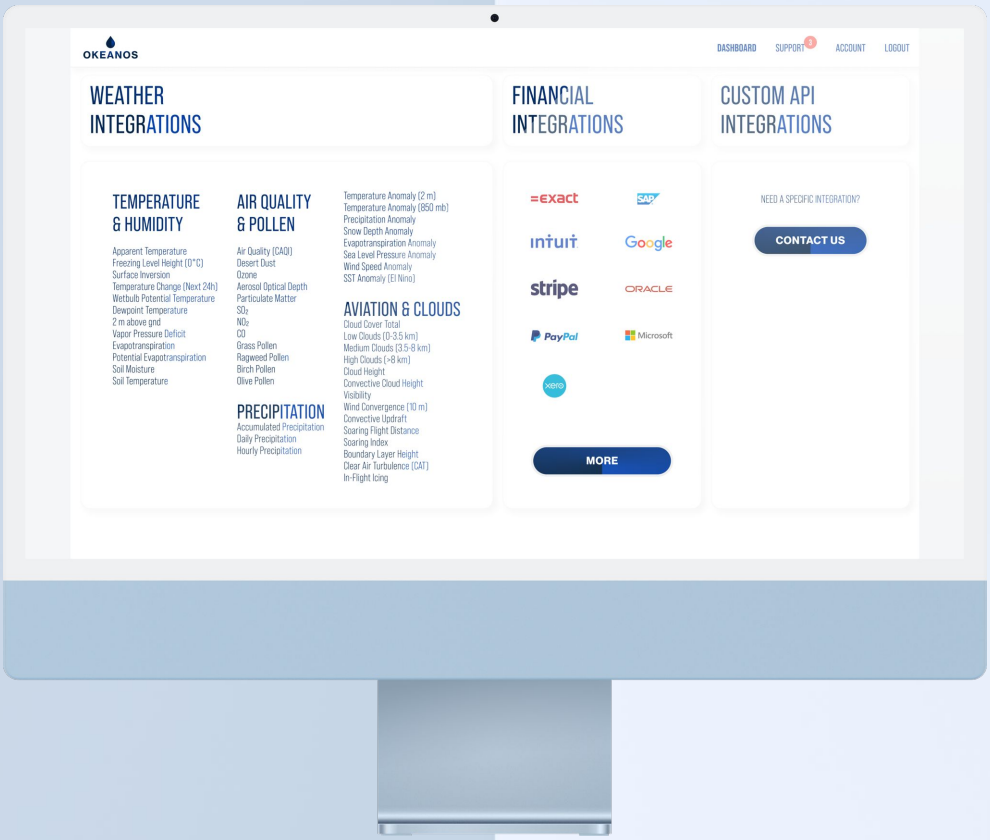
Besides controlling the AWG machines themselves, we can add capabilities to monitor and control the power supplies to the system and control external pumping and storage tanks. This level of control and visibility gives us a unique ability to scale or modular AWG machines into solutions of any size to facilitate large scale supplies for cities and food production.

BILLING AND EXTERNAL DATA INTEGRATIONS

As part of the system monitoring data in regards to the levels of water production across the whole estate or site by site can be accumulated. We can also monitor offtake from the estate as a whole or from individual sites. This data can be used to integrate with an external billing system to keep track of the overall efficiency of the system and enable direct billing of customers.

Other external data sources can be integrated to enhance the performance of the system. We can use external local weather forecast data to give us a projection of water yield. We can monitor and forecast water demand and bring on extra capacity if required.



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